1) $f(g(x))=0$

$f(g(x))=0 \iff g(x)=-1 \vee g(x)=1$

$\ln x=-1 \vee \ln x=1$

$\ln x=e^{-1} \vee \ln x=e^{1}$ (D)

$x=\frac{1}{e} \vee x=e$

2) $\lim_{x\to 3} \frac{f(x)-f(3)}{x^2-3x} = \lim_{x\to 3}\left(\frac{f(x)-f(3)}{x-3}\cdot\frac{1}{x}\right) = f'(3)\cdot\frac{1}{3}$

$= -\sqrt{3}\times\frac{1}{3} = -\frac{\sqrt{3}}{3}$

$f'(3)=\tan\left(\frac{2\pi}{3}\right)=-\sqrt{3}$ (C)

3) a) $cp = {}^{9}C_4\cdot{}^{5}C_2$

b) $cf = {}^{6}C_1\cdot{}^{5}C_2\times 2$

X		X
	X	
X		X

4) $cf = {}^{5}C_2\cdot{}^{7}A_2\cdot{}^{10}A_5 = {}^{5}C_2\cdot{}^{7}C_2\,{}^{2}A_2\cdot{}^{10}A_5$

$= {}^{7}C_2\cdot{}^{5}A_2\cdot{}^{10}A_5$

$= {}^{5}C_2\cdot{}^{10}C_5\cdot 7!$

5) $\vec{n}_\alpha = \vec{r} = (4;0;-1)$ (D)

6) $\lim_{x\to 1/8}\left(e^{-x+a}+\log_5(1-bx)\right)=-\infty$

$e^{-1/8+a}+\log_5\underbrace{\left(1-\frac{b}{8}\right)}_{\to 0}=-\infty$

$1-\frac{b}{8}=0$

$b=8$

$g(-3)=3$

$e^{-(-3)+a}+\log_5(1-8\times(-3))=3$

$e^{3+a}+\log_5 5^2=3 \iff e^{3+a}=1 \iff e^{3+a}=e^0$

$3+a=0$ (B)

$a=-3$

7) $f'(x)=-\sin x+1-\frac{\cancel{2}\cos(2x)}{\cancel{2}}$

$=-\sin x+1-(\cos^2 x-\sin^2 x)$

$=-\sin x+\sin^2 x+\sin^2 x = \sin x(2\sin x-1)$

	0		$\frac{\pi}{6}$		$\frac{5\pi}{6}$		π
f'		−	0	+	0	−	
f	M	↘	m	↗	M	↘	
	1		$\frac{\sqrt{3}}{4}+\frac{\pi}{6}$		$\frac{3\sqrt{3}}{4}+\frac{5\pi}{6}$		$-1+\pi$

$\sin x=0 \vee 2\sin x-1=0$

$\sin x=\frac{1}{2}$

$x=0 \vee x=\frac{\pi}{6} \vee x=\frac{5\pi}{6}$

D: $[\ ;\ ]$

8) $4+\log_a(5^{\ln a}) = 4+\ln a\cdot\log_a 5$

$=\ln e^4+\cancel{\ln a}\cdot\frac{\ln 5}{\cancel{\ln a}} = \ln(e^4\times 5)$ (B)

9.1) $P(1)=120\,e^{-k}$

$P(2)=120\,e^{-2k}$

$P(2)=\frac{1}{2}P(1)$

$\cancel{120}e^{-2k}=\frac{1}{2}\cancel{120}\cdot e^{-k}$

$\frac{e^{-2k}}{e^{-k}}=\frac{1}{2} \iff e^{-k}=e^{\ln\frac{1}{2}}$

$-k=\ln\frac{1}{2} \iff k=-\ln\frac{1}{2}$

$\iff k=\ln\left(\frac{1}{2}\right)^{-1} \iff \underline{k=\ln 2}$

9.2) $P(0)=120$

$P(3)=147$

$TMV_{[0;3]}=\frac{147-120}{3-0}\simeq 35$

10) $P(\bar{A}\cup\bar{B})=0{,}9 \iff P(\overline{A\cap B})=0{,}9$

$\iff 1-P(A\cap B)=0{,}9 \iff P(A\cap B)=0{,}1$

$P(\overline{A\cup B})=0{,}5 \iff 1-P(A\cup B)=0{,}5 \iff P(A\cup B)=0{,}5$

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$0{,}5=2P(B)+P(B)-0{,}1 \iff P(B)=0{,}2$

11.1) $(x;y;z)=\left(\frac{5}{3};2;1\right)+k(2;3;6),\ k\in\mathbb{R}$

2) $\begin{cases} x=\frac{5}{3}+2k \\ y=2+3k \\ z=1+6k \\ 2x+3y+6z+1=0 \end{cases}$ $\begin{cases} = \\ = \\ = \\ \frac{10}{3}+4k+6+9k+6+36k+1=0 \end{cases}$ $\begin{cases} x=1 \\ y=1 \\ z=-1 \\ k=-\frac{1}{3} \end{cases}$

3) $\vec{VC}(-\frac{1}{3};-1;2)$ $\quad \|\vec{VG}\|=\frac{7}{3}$

$V=\frac{\pi\times 1^2\times\frac{7}{3}}{3}=\frac{7\pi}{9}$

12) $\sin\alpha=\frac{\sqrt{3}}{4}$

$\sin^2\alpha+\cos^2\alpha=1$

$\frac{3}{16}+\cos^2\alpha=1$

$\cos^2\alpha=\frac{13}{16} \iff \cos\alpha=\frac{\sqrt{13}}{4}$

$\vec{a}\cdot\vec{b}=\sqrt{7}\times\cancel{4}\times\frac{\sqrt{13}}{\cancel{4}}=\sqrt{39}$

13) $f_n=15+(n-1)\times 5=5n+10$

$f_1=15 \quad r=5$

$G_n=33+(n-1)\times 3=3n+30$

$G_1=33 \quad r=3$

$S_f=S_G \iff \frac{15+5n+10}{\cancel{2}}\times\cancel{n}=\frac{33+3n+30}{\cancel{2}}\times\cancel{n}$

$25+5n=63+3n$

$2n=38$

$n=19$

$f_{19}=105$

$G_{19}=87$